

Motor System Examination



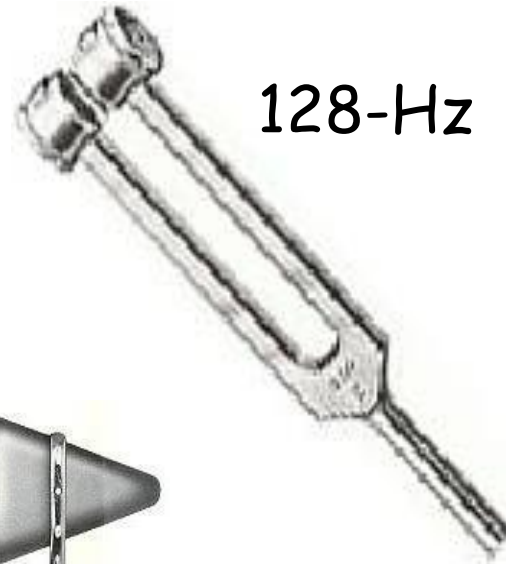
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The Neurological Examination





Neuro Exam Tools



7 categories of the neurological exam

- Mental status
- Cranial nerves
- Motor system
- Reflexes
- Sensory system
- Coordination
- Station and gait

Station is the patient's attitude, posture, or manner of standing. The healthy individual stands erect with her/his head up, chest out, and abdomen in.

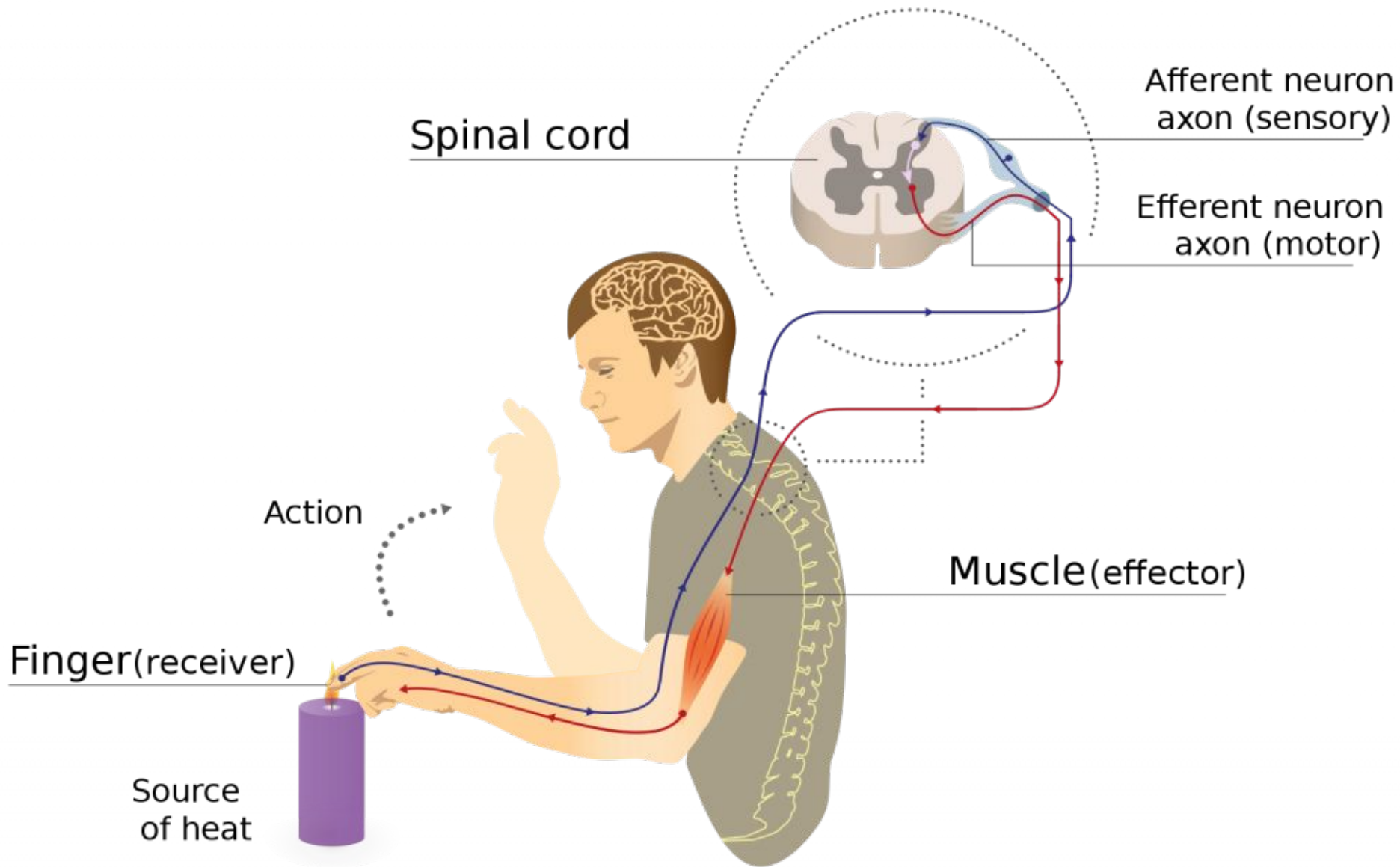
Mental status

7 components of the mental status examination

- **Level of consciousness**
- **Attention**
- **Orientation**
- **Language** — fluency, comprehension, repetition,
- **Memory** -Immediate recal, recent ,remote naming,reading,writing
- **Higher intellectual function**

Intellectual functioning refers to the "general mental ability that includes reasoning, planning, problem solving, abstract thinking, comprehending complex ideas, learning quickly and learning from experience.

ASSESSMENT OF THE MOTOR SYSTEM



ASSESSMENT OF THE MOTOR SYSTEM

1-Inspection and Palpation of the Muscles

2-Assessment of the tones

3-Examination of reflexes

4-Testing movement and power

5-Co-Ordination

Motor exam

Compare left to right, proximal to distal, arms to legs

- Bulk (muscle mass)
- Tone (muscle tension at rest)

Test with passive manipulation

- Strength of the muscle.
- Speed of movement; extraneous movement
- Endurance

Bulk of the muscle

It may be normal, decreased (**atrophy**) or increased (**hypertrophy**).

The wasted and atrophic muscles are small, soft, and flabby.

- Muscle atrophy is seen in neurological disorders, especially in lower motor neuron disease.
- Generalized wasting of muscle occurs in non-neurological conditions like malignancy, diabetes mellitus, thyrotoxicosis, and tuberculosis.
- Hypertrophy occurs due to excessive use of the muscle as in athletes and gymnasts. It can also occur in some myotonic disorders.

Assessment of the tones

It is defined as the state of partial contraction of the muscle.

The decrease in tone is called “**Hypotonia**”.

Increase in tone is named as “**Hypertonia**”.

Hypertonia is a feature of upper motor neuron lesion and it either manifests as Spasticity (clasp knife type) seen in pyramidal tract lesion or Rigidity (seen in all muscles, without any relation to gravity) which occurs in lesions of extra pyramidal tracts.



Assessment of Muscle Tone



Muscle strength OR Muscle Power testing

Definition: Muscle testing is physical examination of muscular strength which helps in diagnosis, progression and treatment of neuromuscular and musculoskeletal disorders



Power of muscles

Decreased muscle strength is called paresis whereas no muscle strength is called paralysis.

Hemiplegia means paralysis of one side of the body, especially of the arm and leg,

Paraplegia means paralysis of both legs,

Monoplegia means paralysis of one limb

Quadriplegia means paralysis of all four limbs.

Gait is the posture of the subject while walking.

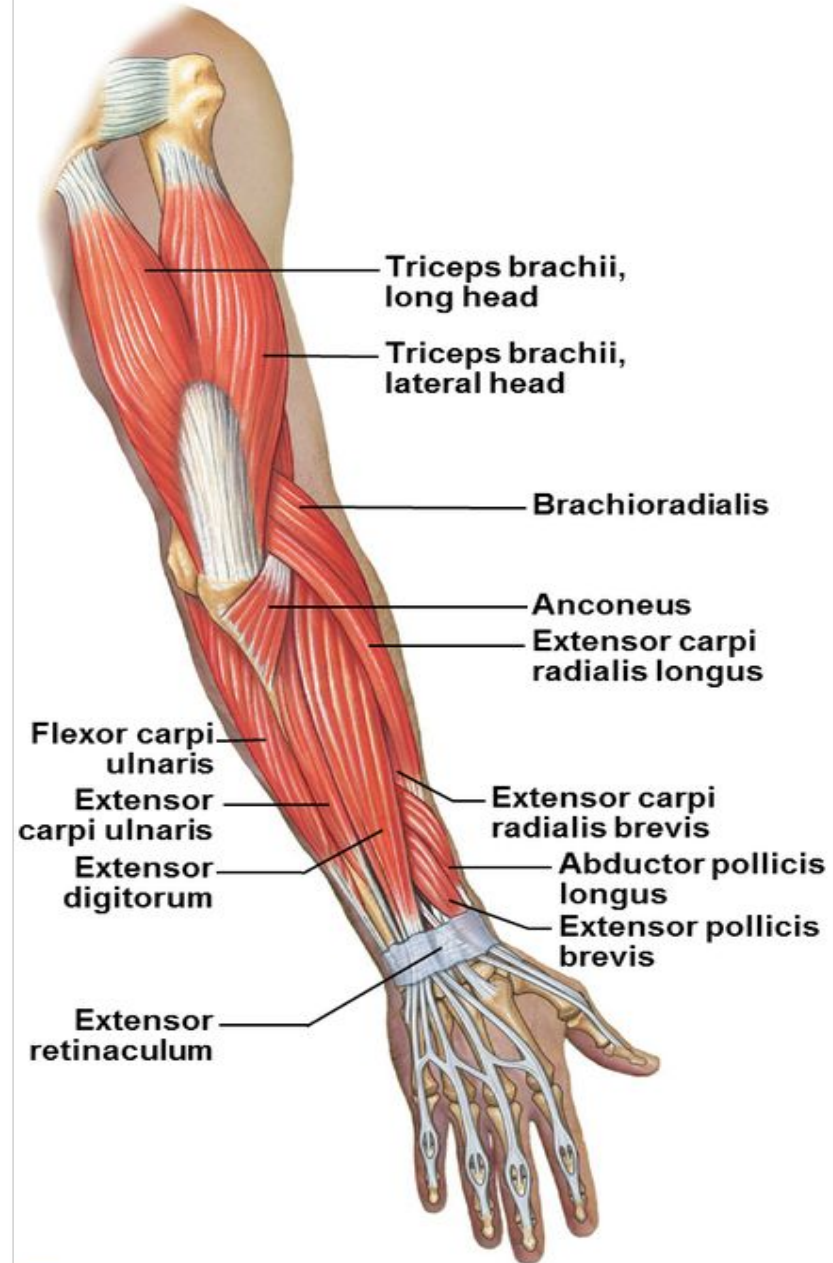
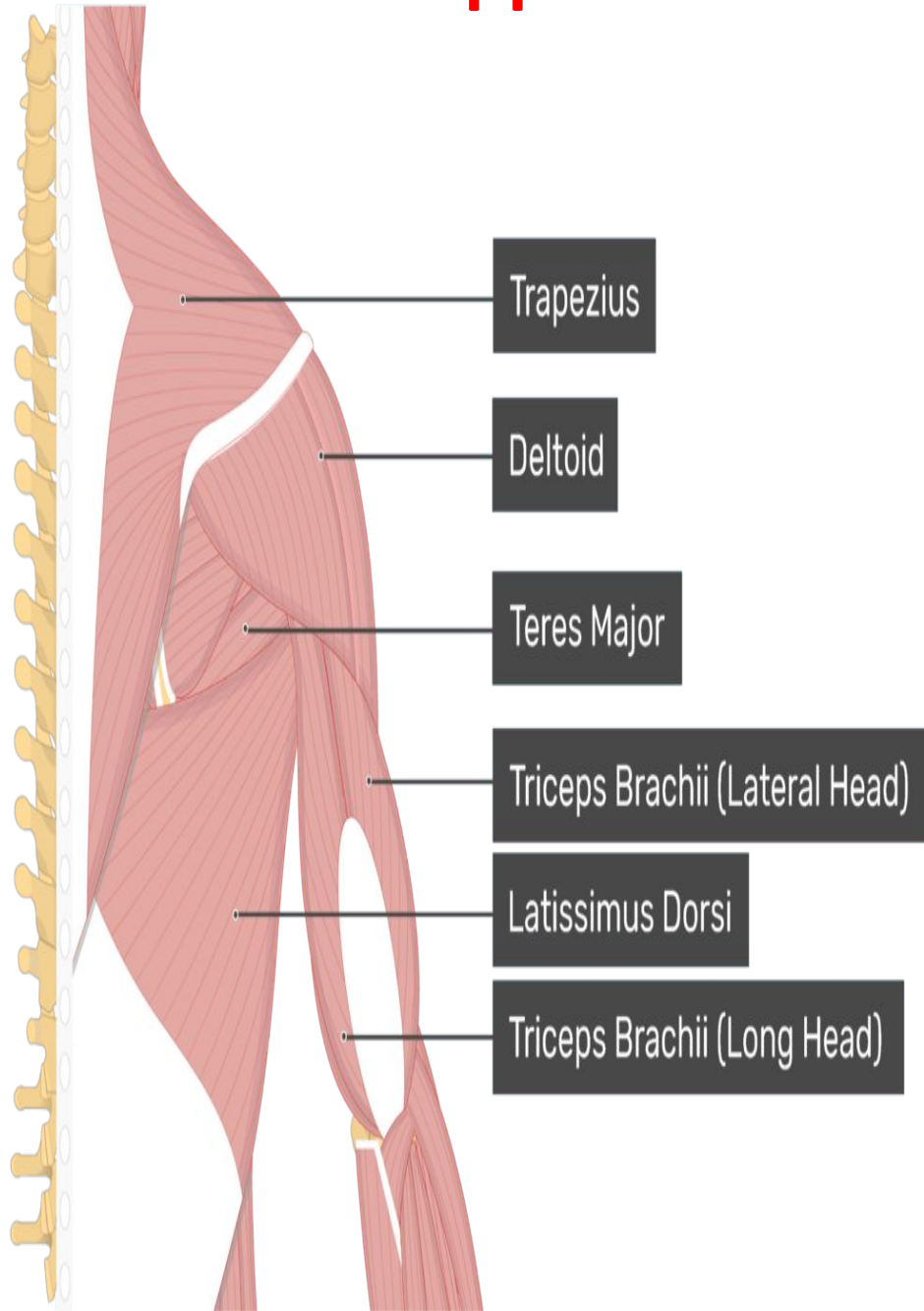
POWER OF MUSCLE:

It is tested by asking the person to bring it into an action against the resistance offered by the doctor. Always compare the muscles of both the sides of the body.

Power is graded as follows

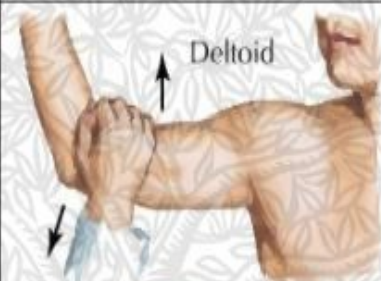
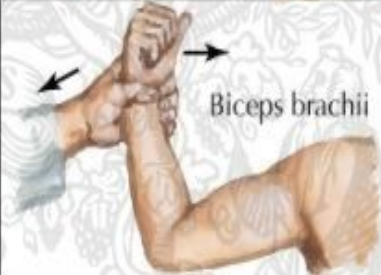
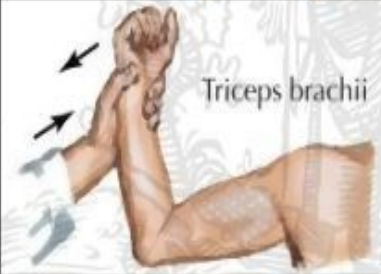
- a. Grade 0 (complete paralysis)**
- b. Grade 1 (flicker of contraction only)**
- c. Grade 2 (patient can move his limb when gravity is excluded)**
- d. Grade 3 (patient can move against the gravity but not against the resistance)**
- e. Grade 4 (patient can move against the resistance but power is less than normal)**
- f. Grade 5 (normal power)**

Upper extremity muscles test



a Posterior view, superficial layer

- **Deltoid**—abduction (elevation) of upper arm
(C5-6, axillary nerve)
- **Biceps**—flexion of forearm at elbow
(C5-6, musculocutaneous nerve)
- **Triceps**—extension of forearm at elbow
(C6-8, radial nerve)

Level	Motor signs (weakness)
C5	 An illustration of a person's right arm abducted at the shoulder. An arrow points to the deltoid muscle on the shoulder, and another arrow points to the axillary nerve at the surgical neck of the humerus.
C6	 An illustration of a person's right arm flexed at the elbow. An arrow points to the biceps brachii muscle on the front of the upper arm.
C7	 An illustration of a person's right arm extended at the elbow. An arrow points to the triceps brachii muscle on the back of the upper arm.
C8	 An illustration of two hands with fingers spread. Arrows point to the interossei muscles in the palm and back of the hand.

- **Extensor carpi radialis**
—dorsiflexion of hand at wrist
(C5-6, radial nerve)

- **Abductor pollicis braves**
—palmar abduction of thumb

(C8-T1, median nerve)
w/ thumb at right angle to palm

- **Interrosei**
—finger abduction (dorsal) & adduction (palmar)
— (C8-T1, ulnar nerve)



Lower extremity muscles test

- **Iliopsoas**—hip flexion
(L1-3, femoral nerve)
- **Quadriceps**—knee extension
(L2-4, femoral nerve)
- **Hamstrings**—knee flexion
(L5-S2, sciatic nerve)
- **Tibialis anterior**—ankle dorsiflexion
(L4-5, deep peroneal nerve)
- **Gastrocnemius/soleus**—ankle plantar flexion
(S1-2, tibial nerve)



Muscular strength and muscular endurance

Muscular strength is the amount of weight you can lift, while muscular endurance is how many times you can move that weight until the muscle gets fatigued.

MUSCULAR STRENGTH vs. MUSCULAR ENDURANCE

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STRENGTH

- Max Amount Of Force A Muscle Can Generate Once
- Heavy Weight
- 1-5 Reps
- 80%+ Of 1 Rep Max
- Ex: Squat 1 Rep Max



ENDURANCE

- Ability To Repeat An Exercise Without Tiring
- Moderate Weight
- 12-20+ Reps
- 30-50% Of 1 Rep Max
- Ex: 50 Push-Ups In A Row



EXAMINATION OF THE REFLEXES

Muscle stretch reflexes

- **Biceps** (C5, C6; musculocutaneous)
- **Triceps** (C6, C7; radial)
- **Patellar** (L2-L4; femoral)
- **Ankle** (S1-S2; tibial)

Biceps reflex

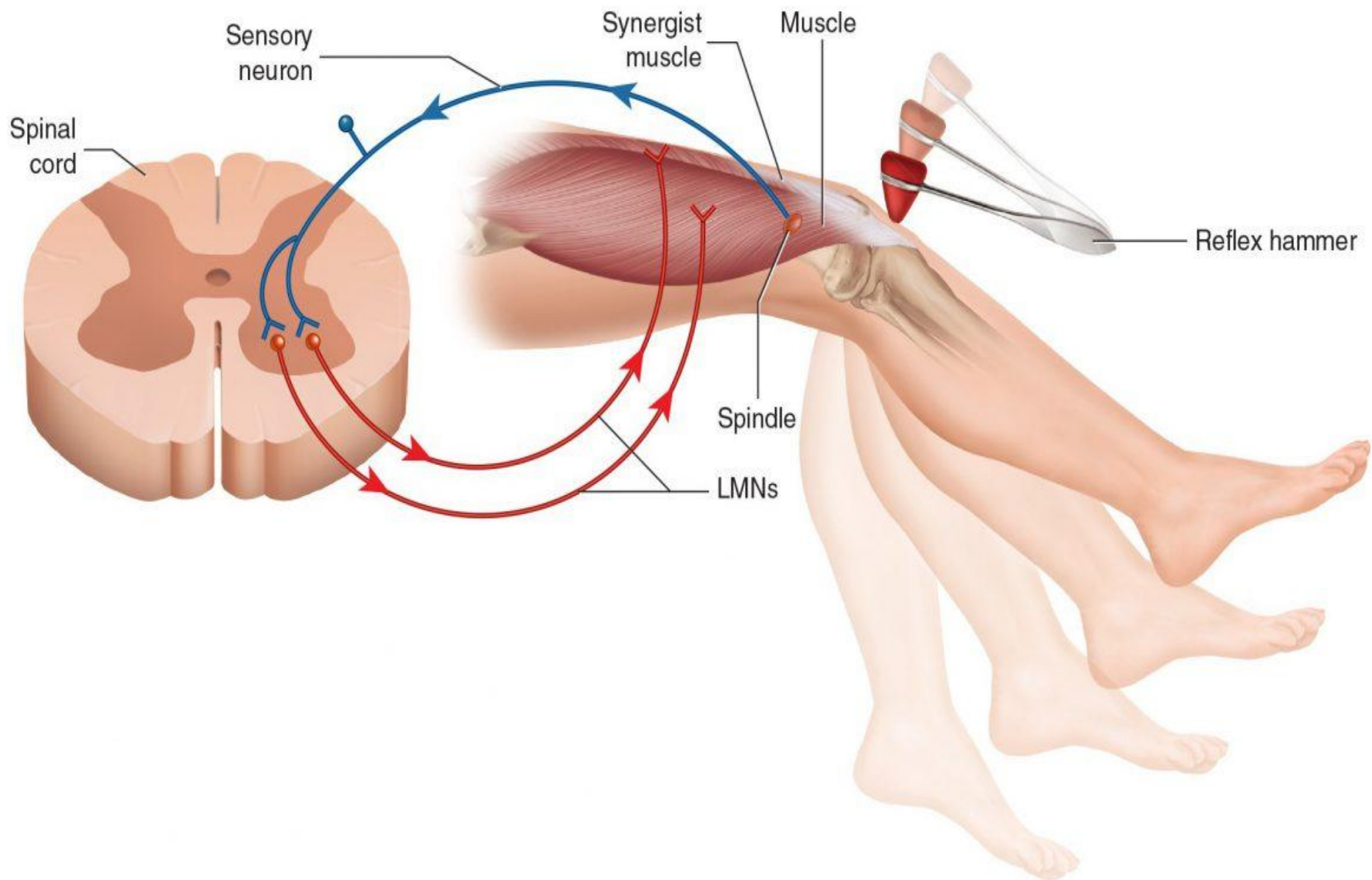


Triceps reflex



Knee Jerk





Ankle Jerk



Reflex grading

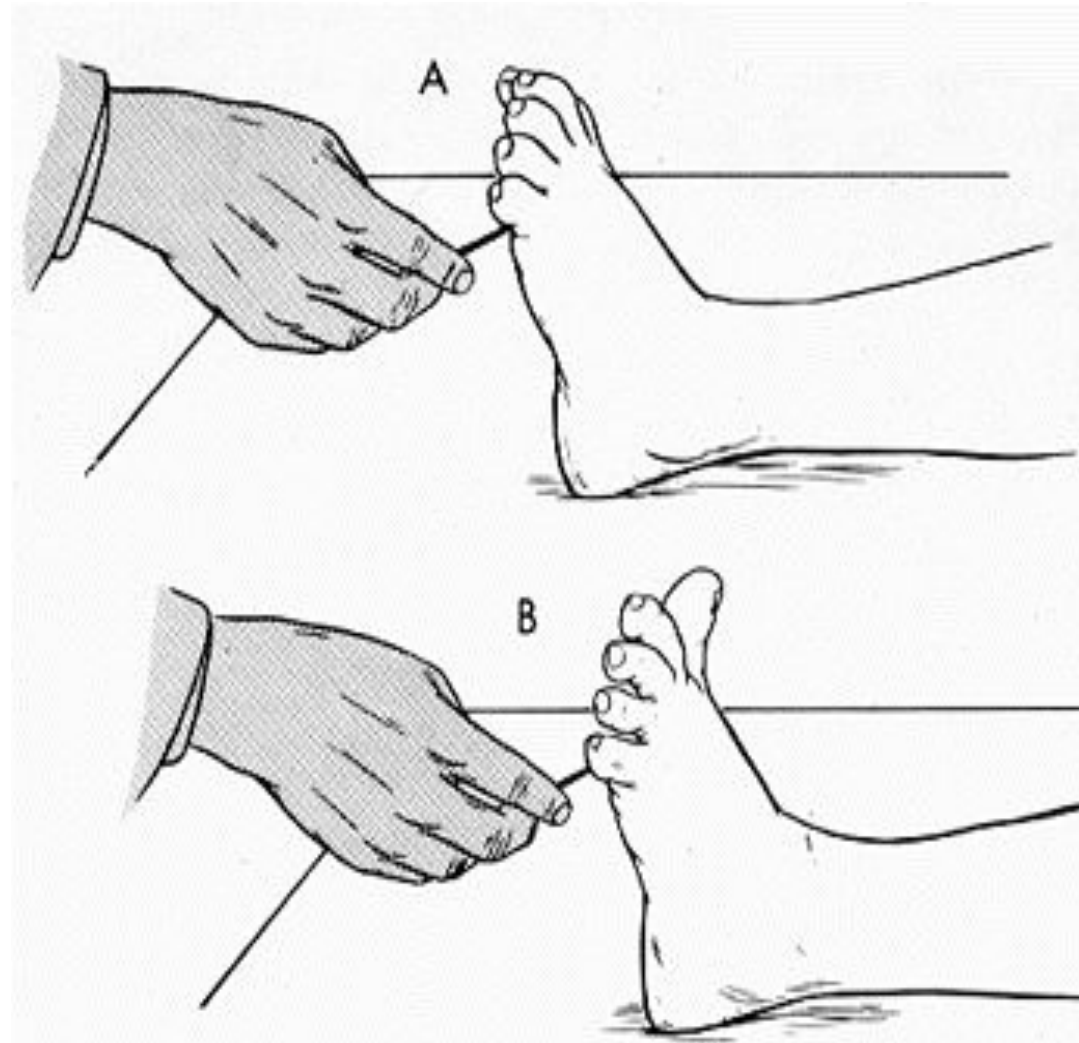
- 1 Absent
- 2 Hypoactive
- 3 Normal
- 4 Brisk/hyperactive
- 5 Markedly hyper active with clonus

Clonus is a rhythmic, oscillating, stretch reflex. The cause of the oscillations are related to lesions in upper motor neurons and is generally accompanied by hyperreflexia.

Clonus is an abnormal reflex response that involves involuntary and rhythmic muscle contractions.



Plantar reflex



Plantar reflex



Babinski Reflex

Negative
Normal



Plantar flexion of all toes

The lateral aspect of the sole is stroked
from the heel to the big toe

Positive reflex is associated with CNS
disorders

Positive
Abnormal



Dorsiflexion of the big toe with the other
toes fanning out
NORMAL FINDING IN INFANTS THROUGH THE
ADOLESCENT PERIOD

Coordination OF MOVEMENTS

It is tested by the following methods.

FOR UPPER LIMB:

- Finger nose test (Person is asked to touch his nose and examiner's finger quickly)
- Finger to finger test (subject is asked to touch the finger of examiner)
- Dysdiadokokinesis. (Person is asked to pronate and supinate the hand quickly to test rapid and alternating movements)
- Writing, buttoning, and threading of a needle.



A



B



Dysdiadochokinesis



- The patient is asked to place one hand on top of the other
- Then is asked to repeatedly and quickly pronate and supinate one hand
- In cerebellar disease the motion is slow and clumsy

FOR LOWER LIMB:

a) To walk along a straight line.

b) Heel -shin test (The subject lying supine is asked to pass the heel of one leg slowly down the shin of the other leg from knee to ankle)

c) Romberg's test. (The subject is asked to stand erect with feet together and eyes closed, ask him to walk straight on the floor)

Cerebellar dysfunction is characterized by ataxia (uncoordinated movements), intention tremor, hypotonia and asthenia (less muscle strength).

FOR LOWER LIMB:

To walk along a straight line.



Heel -shin test

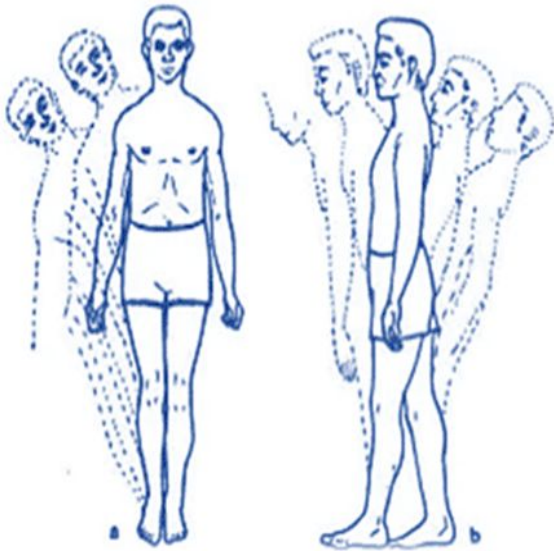
The subject lying supine is asked to pass the heel of one leg slowly down the shin of the other leg from knee to ankle



Romberg's Test



Romberg Test



- Romberg Test is a neurological examination that is used to test for balance and coordination.
- Romberg Test assesses the patient's ability to stand with the feet parallel and together with the eyes open and then closed for 30 s.

TREMORS

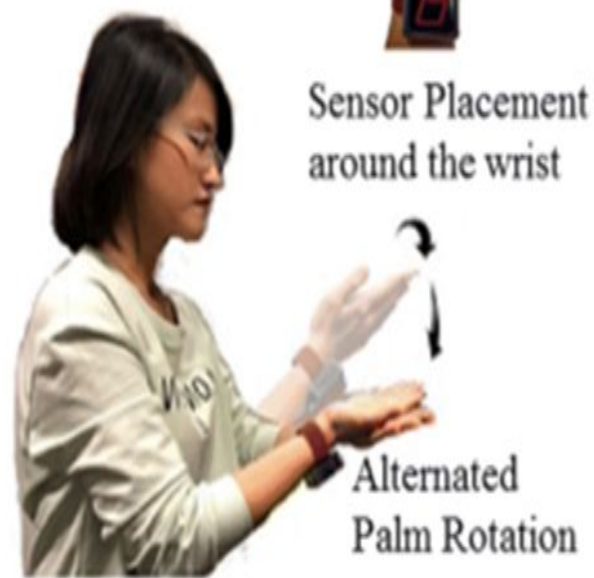
In basal ganglia dysfunction: resting tremors are present

In cerebellar disorders: intention tremors occur (when performing a task)



Upper Limb Tests

Lower Limb Test



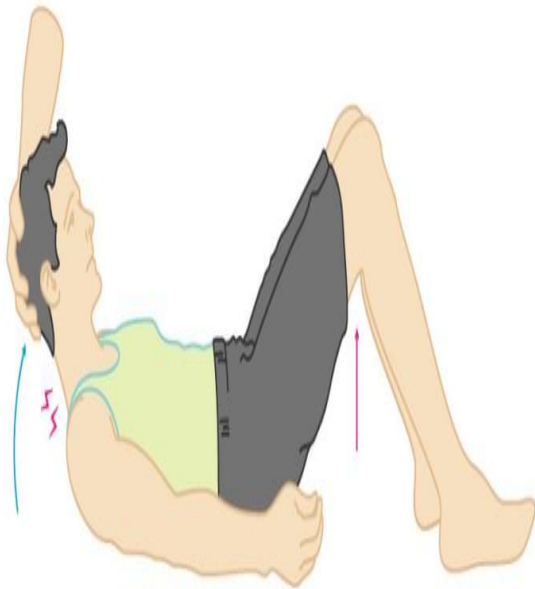
(a) Finger to Nose Test

(b) Dysdiadochokinesia Test

(c) Heel to Shin Test

Meningeal signs

- Neck mobility—look for nuchal rigidity (neck stiffness)



Kernig's Sign



When the patient is lying with the thigh flexed on the abdomen, the leg cannot be completely extended

Relevant Questions

Q-1-How would you characterize the touch receptor distribution pattern in the skin location you tested

The distribution pattern of touch receptors is characterized by high variability across different skin locations, with a higher density of receptors in areas requiring greater tactile acuity, such as the fingertips and palms, and lower density in hairy skin on the arms, legs, and face. This differential distribution allows for fine discrimination of light touch, pressure, and vibration in sensitive areas and a broader detection of stimuli in less sensitive regions.

Q-2-How does the distribution of touch receptors appear to differ between regions?

Touch receptor distribution varies significantly across the body; highly sensitive areas like the fingertips, lips, and tongue have very high densities of receptors, resulting in high tactile resolution and small receptive fields. In contrast, less sensitive regions, such as the back and arms, have lower densities of touch receptors, larger receptive fields, and lower tactile resolution, allowing them to sense touch over broader areas but with less precision. This uneven distribution is often correlated with the presence or absence of hair and the functional requirements of each body part.

Q-3-What area of the skin has the greatest and least ability to differentiate between two points?

The skin areas with the greatest ability to differentiate two points are the fingertips, lips, and tongue, while the back has the least ability. This difference is due to the density of touch receptors, such as Merkel's disks and Meissner's corpuscles, with highly sensitive areas having more receptors and smaller receptive fields, allowing for more precise tactile perception.

Q-4-What do you understand by somatic senses?

Somatic senses, also known as the somatosensory system, are the general sensations that originate from the body, including touch, pressure, vibration, temperature, and pain. These sensations are detected by specialized receptors in the skin and throughout the body and are communicated to the brain via sensory neurons, allowing for perception of the external environment and the body's position in space

Q-5-What is the importance of examining sensory system?

Examining the sensory system is important for diagnosing and locating neurological problems by assessing a patient's ability to sense touch, pain, temperature, and position. This helps determine the extent and severity of sensory impairments and informs subsequent diagnostic steps like imaging or biopsies. In clinical medicine, understanding the sensory system is vital for treating conditions like chronic pain, and in areas like pediatrics, it's crucial for assessing learning and development.

Q-6-Differentiate between primary and cortical sensation-

Primary sensations are basic sensory inputs like touch, pain, temperature, and proprioception, which are transmitted directly from the body to the brain's primary sensory cortex for initial processing. In contrast, cortical sensations are higher-order cognitive abilities that involve more complex processing within the cortex, requiring the brain to interpret and integrate these primary inputs to perform tasks like recognizing an object by touch (stereognosis), identifying letters written on the skin (graphesthesia), or discriminating between two nearby points of touch (two-point discrimination).

Q-7-What is Astereognosis and graphesthesia?

Stereognosis is the ability to recognize an object by touch alone, while graphesthesia is the ability to recognize symbols or numbers written on the skin by touch. Both are forms of tactile sensation and rely on the integrity of the somatosensory cortex. Astereognosis (loss of stereognosis) or graphanesthesia (loss of graphesthesia) can indicate neurological dysfunction, often in the parietal lobe.

Q-8-In hemisection of spinal cord, which sensations are lost on which sides of body?

In hemisection of the spinal cord, known as Brown-Séquard syndrome, there is ipsilateral (same side) loss of motor function, proprioception, vibration, and fine touch, and contralateral (opposite side) loss of pain and temperature sensation below the level of the lesion. For example, a right-sided hemisection results in paralysis and loss of position sense on the right side, while pain and temperature are lost on the left side.